

Resilience, Physical Activity, and Contextual Influences on Youth Happiness During Summer: Insights from Ecological Momentary Assessment



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Introduction

- Resilience is associated with positive affect, and while studies have shown physical activity (PA) predicts positive affect.
- Summer can be a time of more freedom but less support for adolescents
- The present study examined whether resilience moderates the relationship between PA and positive affect (happiness) among adolescents while also exploring contextual factors, social bonding and perceived safety during the summer.

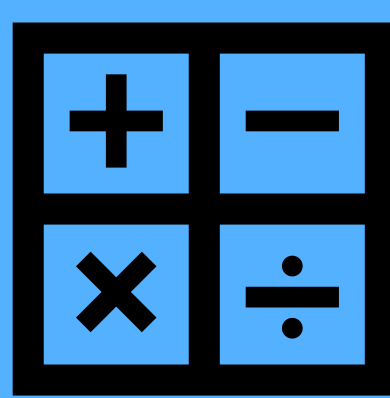
Methods



Youth, ages 11-15, (n=41) participated in a baseline survey, Ecological Momentary Assessment (EMA) questions 6 times a day for seven days, and a wrist-worn Actigraph LEAP



Baseline survey measured resilience via the Connor-Davidson Resilience Scale (CD-RISC 10) and affect was measured via EMA



Multilevel modeling (EMA prompts nested within participants) examined the momentary relationships between PA, positive affect, resilience, and contextual factors (social bonding and perceived safety).

Results

Table 1: Demographics and Key Variables

Variable	n (%) / M (SD)
Age	12.76 (0.86)
Sex	
Male	31 (75.6%)
Female	10 (24.4%)
Resilience (CD-RISC 10)	34.93 (6.53)
Total Prompts Answered	34.51 (5.11)
Overall Completion Rate (%)	82.82 (12.02)
Social Bonding (Average)	2.42 (1.13)
Perceived Safety (Average)	4.39 (0.56)

Mean Positive Affect (Happiness)



3.96 (SD=0.62)

Range 2.5-4.9
Possible Range 1-5



PA in the previous 10 minutes was not significantly associated with momentary affect, and resilience did not moderate this relationship



Resilience showed a significant direct effect on happiness, with more resilient youth reporting higher overall happiness levels



Momentary social bonding and perceived safety were both significantly associated with higher momentary happiness.

Youth who generally experienced higher safety also reported significantly higher happiness

Table 2: Multi-Level Model Results for Positive Affect (Happiness)

Predictor	Coefficient (β)	Std. Error	p-value
Intercept	3.93	0.09	<.001
Within-Person Effects			
PA (10-min deviation)	0.02	0.02	.33
Social bonding	0.05	0.01	<.001
Perceived safety	0.18	0.03	<.001
Between-Person Effects			
PA (person mean)	0.22	0.33	.50
Social bonding (person mean)	0.14	0.07	.07
Perceived safety (person mean)	0.50	0.14	<.001
Resilience (centered)	0.02	0.01	.04
Covariates			
Age (centered)	0.15	0.10	.14
Sex (factor)	0.01	0.20	.98
Interaction Term			
PA × Resilience	-0.004	0.003	.25

Conclusions



Findings suggest that while resilient youth experience higher average happiness, resilience does not moderate the immediate mood effects of short-term PA



Perceptions of safety and social bonding demonstrated consistent effects, underscoring the central role of contextual factors. Fostering environmental and social supports as potential pathways for promoting youth happiness in the summer.

